```
#include<stdio.h>
int check(int day,int month)
{
	if((month==4||month==6||month==9 ||month==11) && day==31)
		return 1;
	else
		return 0;
}
int isleap(int year)
{
	if((year%4==0 && year%100!=0) || year%400==0)
			return 1;
	else
			return 0;

}
int main()
{
	int day,month,year,tomm_day,tomm_month,tomm_year;
	char flag;
	do
	{
		flag='y';
		printf("\nenter the today's date in the form of dd mm yyyy\n")
		scanf("%d%d%d",&day,&month,&year);
		tomm_month=month;
		tomm_year= year;
		if(day<1 || day>31)
		{
			 printf("value of day, not in the range 1...31\n");
			  flag='n';
		}
		if(month<1 || month>12)
		 {
			printf("value of month, not in the range 1....12\n");
			flag='n';
		}
		else if(check(day,month))
		{
			printf("value of day, not in the range day<=30");
			 flag='n';
		}

		if(year<=1812 || year>2013)
		{
```

```
                printf("value of year, not in the range 1812.....2013\n");
                flag='n';
        }
        if(month==2)
        {
                if(isleap(year) && day>29)
                {
                        printf("invalid date input for leap year");
                        flag='n';
                }
                else if(!(isleap(year))&& day>28)
                {
                        printf("invalid date input for not a leap year");
                        flag='n';
                }
        }
}while(flag=='n');


switch (month)
{
        case 1:
        case 3:
        case 5:
        case 7:
        case 8:
        case 10:if(day<31)
                        tomm_day=day+1;
                else
                  {
                        tomm_day=1;
                        tomm_month=month+1;
                  }
                break;
        case 4:
        case 6:
        case 9:
        case 11: if(day<30)
                        tomm_day=day+1;
                 else
                   {
                        tomm_day=1;
                        tomm_month=month+1;
                   }
                break;

        case 12: if(day<31)
                        tomm_day=day+1;
                else
                  {
                        tomm_day=1;
                        tomm_month=1;
```

```
                printf("the next day is out of boundary value of year\n");
                    tomm_year=year+1;
                  }
                 else
                    tomm_year=year+1;
               }
             break;
case 2:
             if(day<28)
                    tomm_day=day+1;
              else if(isleap(year)&& day==28)
                           tomm_day=day+1;
               else if(day==28 || day==29)
                    {
                     tomm_day=1;
                     tomm_month=3;
                    }
              break;

printf("next day is : %d  %d  %d",tomm_day,tomm_month,tomm_year);
return 0;}
```

Test Case Name : Boundary Value Analysis test cases for Next date program

Experiment Number : 11

Test data : Enter the three integer value

Pre-condition : Month 1 to 12 , DAY 1 TO 31 AND YEAR 1812 TO 2013 / we consider one corner for the input space

Brief Description :

	Min	Min +1	Normal	Max -1	Max
Month	1	2	6	11	12
Day	1	2	15	30	31
Year	1812	1813	1912	2012	2013

Next date Output Boundary Value Analysis Cases

Case Id	Description	Input Data			Expected Output			Actual output			Status	Comment
		Month	day	year	Month	day	year	Month	day	year		
1	Enter the min value month, day and year	1	1	1812	1	2	1812					
2	Enter the min+1 value for year and min for month and day	1	1	1813	1	2	1813					
3	Enter the normal value for year and min for month and day	1	1	1912	1	2	1912					
4	Enter the max -1 value for year and min for month and day	1	1	2012	1	2	2012					
5	Enter the max value for year and min for month and day	1	1	2013	1	2	2013					
6	Enter the min+1 value of day and min for month and year	1	2	1812	1	3	1812					

7	Enter the min+1 value for day and year and min for month	1	2	1813	1	3	1813					
8	Enter the min+1 value for day , normal value for year and min value for month	1	2	1912	1	3	1912					
9	Enter the min+1 value for day , max -1 value for year and min value for month	1	2	2012	1	3	2012					
10	Enter the min+1 value for day , max value for year and min value for month	1	2	2013	1	3	2013					
11	Enter the normal value of day and min for year and month	1	15	1812	1	16	1812					
12	Enter the normal value for day and min+1 for year and min for month	1	15	1813	1	16	1813					
13	Enter the normal value for day normal value for year and min value for month	1	15	1912	1	16	1912					
14	Enter the normal value for day , max -1 value for year and min value for month	1	15	2012	1	16	2012					
15	Enter the normal value for day , max value for year and min value for month	1	15	2013	1	16	2013					
16	Enter the max - 1 value of day and min for day and year	1	30	1812	1	31	1812					
17	Enter the max -1 value for day and min for month and min+1 for year	1	30	1813	1	31	1813					
18	Enter the max - 1 value for day , normal value for year and min value for month	1	30	1912	1	31	1912					

19	Enter the max - 1 value for day , max -1 value for year and min value for month	1	30	2012	1	31	2012					
20	Enter the max -1 value for day , max value for year and min value for month	1	30	2013	1	31	2013					
21	Enter the max value of day and min for year and month	1	31	1812	2	1	1812					
22	Enter the max value for day and min for month and min + 1 for year	1	31	1813	2	1	1813					
	Enter the max value for day , normal value for year and min value for month	1	31	1912	2	1	1912					
24	Enter the max value for day , max -1 value for year and min value for month	1	31	2012	2	1	2012					
25	Enter the max value for day , max value for year and min value for month	1	31	2013	2	1	2013					

Next date Output Special value test cases

Case Id	Description	Input Data			Expected Output			Actual output			Status	Comment
		month	day	year	month	day	year	month	day	year		
1	Enter the D1, M1 and Y1 valid cases	12	31	1811	Should display the message value of the year in range 1812..2013							
2	Enter the D1, M1 and Y2 valid cases	12	31	2012	1	1	2013					
3	Enter the D1, M1 and Y3 valid cases	12	31	2013	Should display the message Next is out of boundary 2013							

Test Case Name : Equivalence class test cases for Next date

Experiment Number : 12

Test data : Enter the three integer value

Pre-condition : Month 1 to 12 , DAY 1 TO 31 AND YEAR 1812 TO 2013

Valid Cases

M1 = { month ; 1 ≤ month ≤ 12 }

D1 = { day : 1 ≤ day ≤ 31 }

Y1 = { year : 1812 ≤ year ≤ 2013 }

Invalid cases

M2 = {month : month < 1}

M3 = {month : month > 12}

D2 = {day : day < 1}

D3 = {day : day > 31}

Y2 = {year : year < 1812}

Y3 = {year : year > 2013}

Next date Output Equivalence Class Testing

(Weak and Strong Normal Equivalence Class)

Case Id	Description	Input Data			Expected Output			Actual output			Status	Comment
		month	day	year	month	day	year	month	day	year		
WN1,SN1	Enter the M1, D1 and Y1 valid cases	6	15	1912	6	16	1912					

(Weak Robustness Equivalence Class)

Case Id	Description	Input Data			Expected Output			Actual output			Status	Comment
		month	day	year	month	day	year	month	day	year		
WR1	Enter the M1, D1 and Y1 cases	6	15	1912	6	16	1912					
WR2	Enter the M2 , D1 and Y1 cases	-1	15	1912	Should display the message value of the month not in the range 1..12							
WR3	Enter the M3 ,D1 and Y1 cases	13	15	1912	Should display the message value of the month not in the range 1..12							
WR4	Enter the M1, D2 and Y1 cases	6	-1	1912	Should display the message value of the day not in the range 1..31							
WR5	Enter the M1, D3 and Y1 cases	6	32	1912	Should display the message value of the day not in the range 1..31							
WR6	Enter the M1, D1 and Y2 cases	6	15	1811	Should display the message value of the year not in the range 1812..2013							
WR7	Enter the M1, D1 and Y3 cases	6	15	2014	Should display the message value of the year not in the range 1812..2013							

(Strong Robustness Equivalence Class)

Case Id	Description	Input Data			Expected Output	Actual Output	Status	Comment
		month	day	year				
SR1	Enter the M2 , D1 and Y1 cases	-1	15	1912	Should display the message value of the month not in the range 1..12			
SR2	Enter the M1, D2 and Y1 cases	6	-1	1912	Should display the message value of the day not in the range 1..31			
SR3	Enter the M1, D1 and Y2 cases	6	15	1811	Should display the message value of the year not in the range 1812..2013			
SR4	Enter the M2 , D2 and Y1 cases	-1	-1	1912	(i)Should display the message value of the month in range 1..12			
					(ii) Should display the message value of the day in range 1..31			
SR5	Enter the M1, D2 and Y2 cases	6	-1	1811	(i) Should display the message value of the day in range 1..31			
					(ii) Should display the message value of the year in range 1812..2013			
SR6	Enter the M2, D1 and Y2 cases	-1	15	1811	(i) Should display the message value of the month in range 1..12			
					(ii) Should display the message value of the year in range 1812..2013			
SR7	Enter the M2, D2 and Y2 cases	-1	-1	1811	(i)Should display the message value of the month in range 1..12			
					(ii) Should display the message value of the day in range 1..31			
					(iii) Should display the message value of the year in range 1812..2013			

Some addition equivalence Boundary checking

Case Id	Description	Input Data			Expected Output			Actual Output			Status	Comment
		day	month	year	day	month	year	day	month	year		
1	Enter the D1, M1 and Y1 valid cases	31	12	1811	Should display the message value of the year in range 1812..2013							
2	Enter the D1, M1 and Y2 valid cases	31	12	2012	1	1	2013					
3	Enter the D1, M1 and Y3 valid cases	31	12	2013	Should display the message Next is out of boundary 2013							